

Harsh Kashyap

Bengaluru, India | +91 80516 25669

harsh.kashyap2001@gmail.com | linkedin.com/in/harsh-kashyap | github.com/Harsh23Kashyap | harshkashyapportfolio.netlify.app

OBJECTIVE

Driven **Software Development Engineer** at **Amazon** with 3+ years building AI platforms and cellular-protocol systems at Apple. Seeking an **MS in Computer Science** focused on **AI/ML and large-scale distributed systems**, with the goal of growing into a **Machine Learning Infrastructure Engineer** building reliable, high-performance systems for large-scale AI and data platforms.

EDUCATION

Thapar Institute of Engineering & Technology

Aug 2019 – Jun 2023

B.E., Computer Science; Minor: Data Science Patiala, India **CGPA: 9.71/10**. Coursework: **Machine Learning, Data Science**, Operating Systems, DBMS, Computer Networks, Computer Architecture, Data Structures & Algorithms, OOP, High- and Low-Level System Design.

PROFESSIONAL EXPERIENCE

Amazon

Jul 2026 – Present

Software Development Engineer II

Bengaluru, India

- Built **Unified Compliance Intelligence**, a compliance-assistance platform for **Amazon Marketplace** that flags non-compliant product listings and gives sellers clear explanations of violations, affected products, and required documentation updates.
- Built backend services with **Java, Coral, ECS Fargate**, and **Amazon Bedrock**, using **Claude/Sonnet** workflows to translate policy and product signals into **auditable compliance guidance**.

Apple

Dec 2023 – Jun 2026

Software Engineer II

Bengaluru, India

- Built and shipped a native **macOS SwiftUI** application for **LTE/5G-NR protocol-message authoring**, replacing fragmented ASN.1 and terminal workflows for **600+ engineers**, driving **80% adoption** and saving about **\$720K annually**.
- Designed a **protocol-aware pre-validation framework** for interdependent LTE/5G-NR message fields, cutting debug time from **20 minutes to under 2 minutes** and improving incident MTTR by **40%** through automated fix recommendations.
- Architected a **production AI engineering platform** with **3GPP-grounded hybrid retrieval, multi-agent orchestration, Jenkins CI/CD**, and **Docker** for protocol engineering, log analysis, and incident triage across **8 teams and 700+ monthly incidents**.
- Led end-to-end bring-up of a **cellular conformance-testing lab** with iPhone and USRP X410/B210 setups, using **Python automation** to reduce setup from **2–3 weeks to under 2 days** and make deployments repeatable.
- Led the **AI/ML subsystem** of a greenfield **LTE Layer-2 code-generation** initiative across MAC, RLC, and PDCP, coordinating **56 agents** and domain experts to generate and validate **100K+ lines of C/C++**.
- Built a **3GPP knowledge platform** that improved retrieval accuracy **32%** over a leading commercial AI system, surfaced **26+ specification-reference errors**, and mentored **2 interns** on an AI workspace optimizer.

PayPal

Jul 2023 – Oct 2023

Software Engineer

Bengaluru, India

- Improved the **Global Merchant Lending admin tool**, driving about **13% audit-efficiency gain** by automating manual workflows with **React, Node.js**, and internal PayPal frameworks in a CECL-compliance context.
- Delivered the work end-to-end in a small pod, coordinating with **Finance, Risk, and Compliance** stakeholders to meet a regulatory deadline with **audit-trail coverage**.

STARTUP EXPERIENCE

Neatlogs

Open-source observability platform for AI agents and LLM applications

Python, TypeScript, Go, OpenTelemetry

- Hardened **TypeScript SDK** transport behavior to prevent duplicate API writes after network failures; validated **5,350 fault-injected mutations with zero replays** across packaged **Node.js** consumers.
- Qualified **Python and TypeScript SDKs**, instrumentation Wizard, standalone Skills, and setup docs; built deterministic harnesses for **streaming**, tool calls, workflow hierarchy, errors, cancellation, privacy masking, concurrency, lifecycle management, endpoint isolation, and trace-replay diagnostics.

INTERNSHIP EXPERIENCE

Apple

Jan 2023 – Jun 2023

Software Engineering Intern

Hyderabad, India

- Built an internal **observability SPA** with **React, Tailwind CSS, Spring Boot**, and **Splunk**, adding real-time dashboards that automated recurring health checks and cut support turnaround by **83%**.
- Designed and deployed an **XGBoost SLA-resolution predictor** with feature engineering, reaching **80%+ accuracy** and reducing SLA breaches by **25%** across IS&T incidents.

Tata Img

Oct 2022 – Nov 2022

Engineering Intern

Gurugram, India

- Engineered asynchronous task scheduling with **Asyncio, Celery**, and **RabbitMQ**, offloading long-running jobs and streamlining operations for **3,000+ employees**, with projected benefit for **4M+ customers**.

Cisco

Jun 2022 – Jul 2022

Technical Undergraduate Intern

Bengaluru, India

- Built **Python** pipelines for **SD-WAN telemetry** and surfaced QoS metrics such as latency, jitter, and packet loss on an internal dashboard, improving network-ops triage efficiency by **23.2%**.

Licious

Jan 2022 – May 2022

Software Developer Intern

Bengaluru, India

- Migrated sections of the application from **Swift** to **Flutter**, refactoring **5,000+ lines**, preserving feature parity, and reducing estimated engineering effort by about **30%**.

YASH Technologies

Jul 2021 – Sep 2021

Software Engineering Intern

Hyderabad, India

- Built an **Employee Management System** portal with **Java, Spring Boot, SQL**, and **REST APIs** for employee records, attendance, and leave management for **6,000+ daily users**.

RESEARCH EXPERIENCE

University of Edinburgh

Prof. Boris Grot

- Built an explainable **pre-execution runtime cost model** for **Apache DataFusion** and **Parquet** query planning, capturing operator-specific scaling across **704 query/dataset pairs** and enabling cost-aware serverless execution.

New York University

Prof. Dennis Shasha

- Built **CustomNerd/NerdEngine**, a config-first no-code framework for domain-specific **RAG assistants** with a **9-stage pipeline**, curated retrieval, prompt refinement, citation generation, saved-state deployment, and **6+ domain variants**.

- Re-architected the backend to serverless **AWS Lambda/S3** with caching and provenance safeguards, reducing cost by about **70%**, scaling from one lab to **10+ research labs**, and benchmarking CloudNerd on a **120-question retrieval benchmark**.

Washington State University

Prof. Parteek Kumar

- Co-designed **JobMatch**, a role-separated **multi-agent system** for explainable job-candidate matching, combining semantic search, a **6-signal composite ranker**, explanation panels, and a CI regression gate.

University of California, Los Angeles

Prof. Lixia Zhang; Named Data Networking

- Contributed **certificate-revocation and trust-validation support** to the public **Go** implementation of **Named Data Networking**, adding Revoke/IsRevoked APIs, process-local revocation records, and trust-schema validation.

University of British Columbia

Prof. L.A.K.S.; ScientificPerplexity

- Contributing to **ScientificPerplexity**, an **LLM-driven scientific-literature query system**; refactored the LLM layer into a provider-agnostic client across **16 providers** with typed configuration and cleaner provider boundaries.

PUBLICATIONS

- CustomNerd: A Framework and Tool for Fast Deployment of Production-Ready Expertise-Based QA Systems**. MDPI Electronics. **First author**; accepted for publication.
- JobMatch: An Agentic Multi-Role Platform for Explainable Job-Candidate Matching**. Autonomous Agents and Multi-Agent Systems. **Co-first author**; in preparation.
- InvestorNerd and NewsNerd**. Domain-specific grounded-AI assistants at NYU. Contributing author; in preparation.

PROJECTS & OPEN SOURCE

- ENCELA – self-hosted personal-cloud platform**: building a multi-service backend with **100 GB Drive storage**, real-time chat, an office suite, Git repository hosting, and a companion Android client.
- Dify & RAGFlow**: contributed reliability improvements to leading open-source **AI/RAG platforms** across backend APIs, document-processing pipelines, search correctness, retry behavior, test coverage, stored-XSS sanitization, sandbox hardening, dependency updates, and new model support.
- Resume Matcher**: built an AI resume-job matching system with **FastAPI**, multi-modal similarity scoring, agentic negotiation, structured ranking explanations, and automated validation for matching quality.
- Research Mailer**: built a multi-mode outreach automation pipeline with template rotation, pacing controls, browser automation, and delivery safeguards for research-internship communication workflows.
- AI Terminal & CodeClue**: built developer-facing AI tools for command assistance, coding hints, smart auto-completion, process/port utilities, and natural-language workflow support.
- Placement and analytics dashboards**: built **Streamlit** and web dashboards for placement records, company analytics, filtering, trend exploration, and outcome visualization.

SELECTED TECHNICAL STRENGTHS

- Comfortable owning **ambiguous systems work end-to-end**: requirement discovery, backend design, UI workflow design, validation harnesses, stakeholder iteration, launch support, and operational debugging.
- Strong **applied-AI engineering** focus across RAG evaluation, provider abstraction, agent orchestration, failure-mode testing, trace observability, ranking systems, and human-readable explanations.
- Built and maintained learning resources across **core CS, system design, DSA, GRE preparation, and interview workflows**; uses notes and tooling to teach concepts clearly and mentor other students and engineers.

ADDITIONAL ENGINEERING ARTIFACTS

- Maintains a public portfolio of engineering notes covering **operating systems, DBMS, computer networks, OOP, concurrency, system design, DSA patterns, and interview preparation**.
- Built practical automation utilities for **browser workflows**, ticket monitoring, email workflows, podcast organization, personal finance tracking, weather views, and mobile-style health calculators.
- Built public web experiences and deployment-backed prototypes with **TypeScript, React, Streamlit, Selenium, Lovable, GitHub-backed publishing, and responsive UI patterns**.
- Uses project writeups, source links, certificates, and mentoring material to turn implementation work into reusable proof for recruiters, collaborators, students, and engineering reviewers.

TECHNICAL SKILLS

Languages	Python, Java, C/C++, Go, JavaScript/TypeScript, SQL
Backend & Web	Spring Boot, FastAPI , Flask, Node.js, REST APIs, React, Tailwind CSS, SwiftUI, Flutter
Cloud & Infra	AWS EC2, S3, Lambda , DynamoDB, SES, Docker, Jenkins, CI/CD, Cloudflare, Redis, RabbitMQ
Data & AI	RAG, multi-agent systems , FAISS, Chroma, Qdrant, SentenceTransformers, XGBoost, TensorFlow/Keras, Apache DataFusion, Parquet
Tools	Git, GitHub, pytest, Postman, Splunk, Shell scripting

CERTIFICATIONS

- Machine Learning Specialization** and **Deep Learning Specialization**, DeepLearning.AI / Andrew Ng; Cisco CCNA; Splunk; SwiftUI.
- Apple experience certificate**; additional professional, internship, and project certificates maintained in the portfolio certificate archive.
- Continued self-directed learning across **distributed systems, ML infrastructure, observability, retrieval systems, cloud architecture, and engineering leadership**.

AWARDS, LEADERSHIP & COMMUNITY

- TIET Merit Scholarship** recipient, ranking in the **top 0.5%** of the cohort across all four undergraduate years.
- Ranked **89th out of 100,000 students** in the AICTE IICC coding competition; featured in Apple's Innovation Deck for an Apple Pay rewards concept selected from **350+ submissions**.
- LinkedIn Top Voice** in Computer Science and Software Engineering with about **40K followers**; solved **1,700+ LeetCode problems** with contest rating around **1,693**.
- Vice Lead Ambassador, Thapar Counselling Cell: led about **100 ambassadors** and a campus-wide mental-health awareness campaign; Placement Coordinator for **2,000+ students and 30+ companies**.
- Built student-facing and community-facing systems including an IDIA volunteer application, counselling-cell outreach workflows, and placement coordination artifacts.
- Community: DiiDi mentoring for education of underprivileged girls; IDIA member/volunteer app; active mentoring through Topmate with **450+ bookings and 50+ testimonials**. Languages: English and Hindi.